COLLEGE OF MEDICINE

Monkey pox and chicken pox

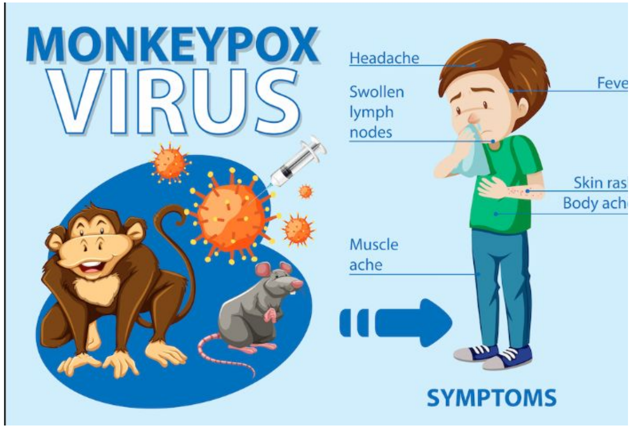
Prepared by :
Varin Tofiq Abbas
Infectious disease control trainee

Learning outcome :

- Understand the Virology and Classification
- Compare Monkeypox and Chickenpox
- Explain Epidemiology & History
- Describe Transmission Routes
- Identify Clinical Features and Stages
- Perform Differential Diagnosis

Introduction

- Monkeypox is a viral zoonotic disease, caused by monkeypox virus, that occurs primarily in tropical rainforest areas of central and west Africa and is occasionally export to other regions.
- It is an enveloped double-stranded DNA virus.
- It has symptoms like those seen in the past in smallpox, but less sever.



Causative Agents: Monkeypox is caused by an orthopoxvirus related to smallpox, while chickenpox is caused by a type of herpes virus.

Rash: Monkeypox sores are usually larger and deeper in the skin, while chickenpox sores arrive in waves.

Distribution: Chickenpox affects only one part of the body, while monkeypox can affect the entire body.

Distinctive Feature: Swollen lymph nodes are a distinctive feature of monkeypox, which are not present in chickenpox

Enveloped double stranded DNA virus with a genome size of around 190 kb

Family :Poxviridae.

Genus : Ortho poxvirus.

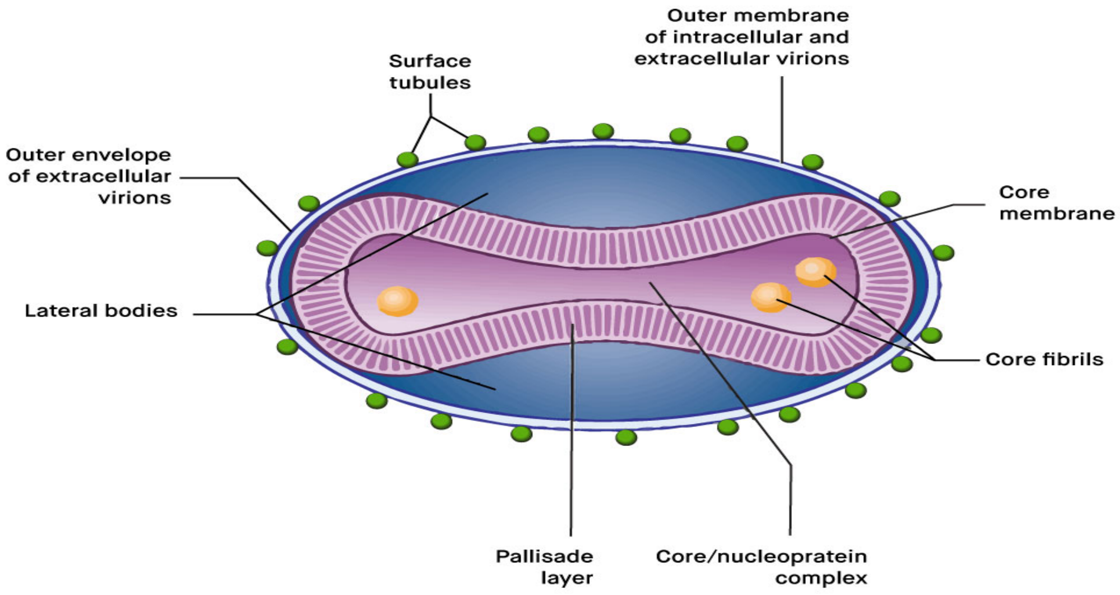


Figure1.Structure of Monkeypox Virus[1]

COLLEGE OF MEDICINE

History

- Monkeypox- Initial discovery in monkeys in a Danish laboratory in 1958.
- First human case- 9 years old boy in the democratic republic of the Congo in 1970.
- It has mortality rate of between one in 100 people, with most deaths occurring in younger age group.

Monkeypox cases in non-endemic countries reported to WHO between 13 to 21 May 2022

Country	Confirmed	Suspected
Australia	1-5	-
Belgium	1-5	1-5
Canada	1-5	11-20
France	1-5	1-5
Germany	1-5	-
Italy	1-5	-
Netherlands	1-5	-
Portugal	21-30	-
Spain	21-30	6-10
Sweden	1-5	-
United Kingdom	21-30	-
United States of America	1-5	-
Total	**92**	**28**

Mode of transmission

very impo

Virus enters body through broken skin, respiratory tract or mucous membrane (eye, nose and mouth)

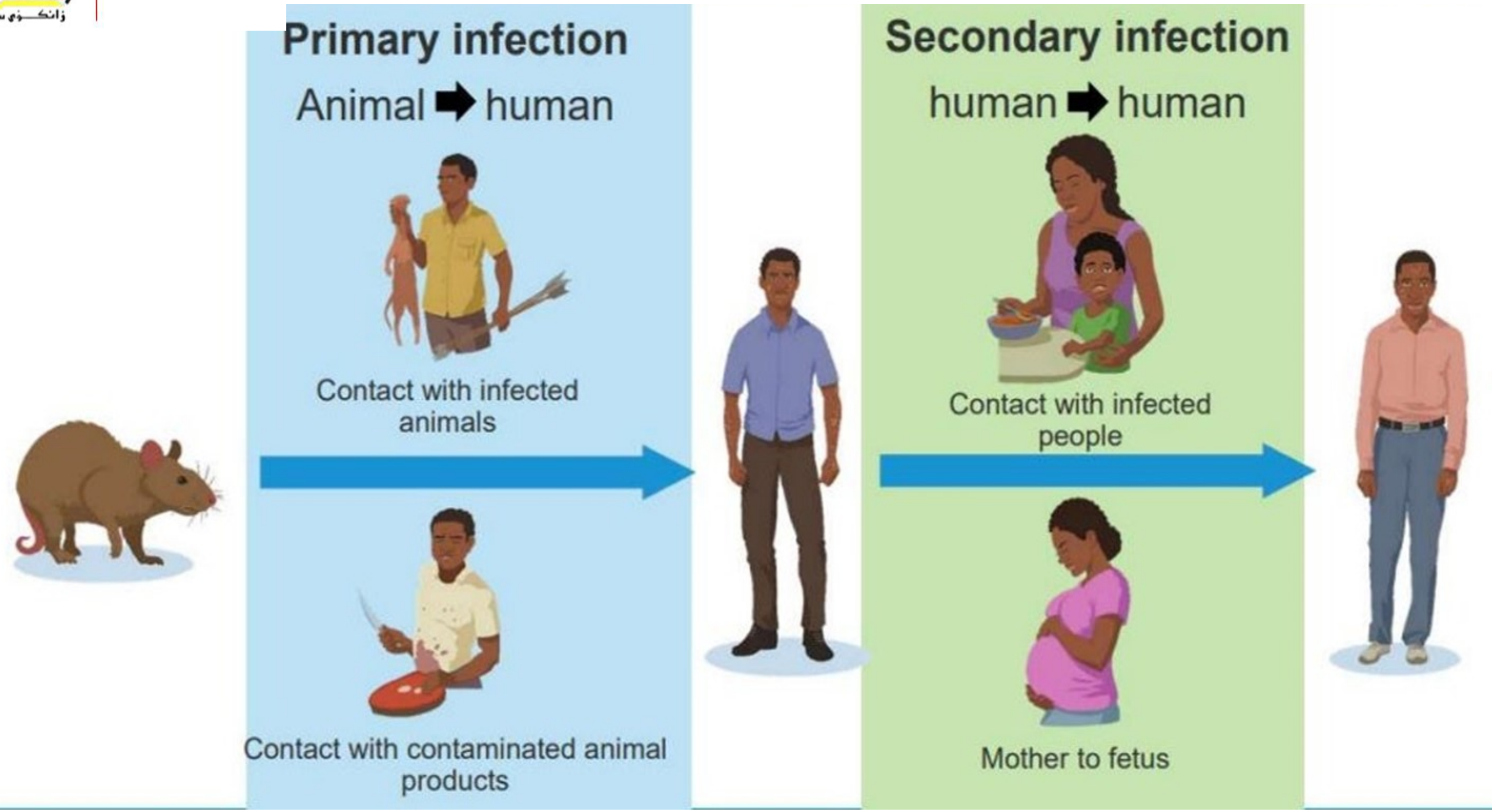
Animal to human

- Bites , scratches and bushmeat.

Human to human

* any blood , body fluid, lesion always wear gloves

- **Direct contact:** Close contact with lesions, body fluids and respiratory droplets.
- **Indirect contact:** contaminated materials such as bedding.


Primary infection
Animal ➡ human
Contact with infected animals
Contact with contaminated animal products
Secondary infection
human ➡ human
Contact with infected people
Mother to fetus

Clinical feature

Incubation period

- 6 to13 days but can range from 5 to 21 days.
- No symptoms .
- Virus present in bloodstream (viremia) at the end of the incubation period.

Invasion period

- Lasts between 1-4 days.
- Characterized by fever, intense headache, lymphadenopathy, back pain, myalgia and intense asthenia. lethargy
- Lymphadenopathy is a distinctive feature of monkeypox compared to other diseases that may initially appear similar(chickenpox and measles).

Period of rash

- Within 1-3 days of appearance of fever.
- More on the face and extremities rather than on the trunk.
- It affects face (95%), and palms of the hands and soles of the feet (75%) ,also affected oral mucous membrane (70%), genitalia (30%), and conjunctivae and cornea (20%).

- The rash evolves from macules to papules, vesicles, pustules and crusts which dry up and fall off.
- Monkeypox is usually a self-limited disease with the symptoms lasting from 2 to 4 weeks.

Differential diagnosis

- Smallpox
- Chickenpox
- Measles
- Bacterial skin infection (especially staphylococcal aureus)
- Scabies
- Syphilis
- Medication-associated allergies

Smallpox

Monkeypox

COLLEGE OF MEDICINE

Laboratory diagnosis

- **Confirmatory test:** Polymerase chain reaction (PCR) is the preferred laboratory test.
- Confirmation of MPXV infection is based on NAAT (Nucleic acid amplification test), using real-time or conventional polymerase chain reaction (PCR), for detection of unique sequence of viral DNA.
- PCR can be used alone or in combination with sequencing.
- For PCR, a viral swab taken from one or more vesicles or crusts, store in dry, sterile tube and keep cold

- Blood/serum-often inconclusive:
- Short duration of viremia.
- Timing of specimen.
- Sodium and potassium test.
- Protein-especially albumin.
- Routine examination of bacterial and mycotic cultures and sensitivity.

COLLEGE OF MEDICINE

Other methods

- **<u>Electron microscopy:</u>**
- Can be used, this method is not routinely used for the diagnosis of poxviruses.
- **<u>Viral culture:</u>**
- Virus isolation is not recommended as a routine diagnostic procedure.

•Infection prevention and control

prevention most imp

COLLEGE OF MEDICINE

Infection prevention and control

1. Education of health care workers
2. Isolation of patient. → separate room, or cohort (all patient in the room have same disease
3. Applying standard precaution e.g Proper hand hygiene and cleaning procedures.
4. Household disinfection.

- **Public health educational messages should focus on the followings:**

 Raise awareness of the risk factors.

 Reduce exposure to the virus.

 Surveillance measures.

 Rapid identification of new cases.

 Limit direct exposure to the infected animals and patients.

Isolation of patients

- Persons with extensive lesions should be isolated in a separate room. with hand wash facility since alcohol rub never replace hand wash
- Household members should limit contact with the infected person.
- Infected people should also avoid contact with animals.

COLLEGE OF MEDICINE

Use of personnel protective equipment

- Persons with monkeypox should wear a surgical mask.
- Disposable gloves should be worn for direct contact with lesions.
- Skin lesions should be covered to the best extent possible (Long sleeves and long pants) to minimize risk of contact with others.
- Contain and dispose of contaminated waste (Dressing and bandages) should be done according to biomedical waste disposal. yellow bag

COLLEGE OF MEDICINE

Proper hand hygiene and clinical procedure

- Hand hygiene-Hand washing with soap and water or use of an alcohol-based hand rub should be done.
- Hand hygiene is to be performed after touching lesion material, clothing, linens and environmental surfaces.

Household infection

- Laundry: linen of infected cases should be washed separately to avoid direct contact with contaminated material.
- Soiled dishes and utensils should be washed with warm water and soap.
- Contaminated surfaces should be cleaned and disinfected. standard household cleaning/disinfectant may be used.

ex: chlorine

COLLEGE OF MEDICINE

Vaccination

- Data from Africa suggests that smallpox vaccine is at least 85% effective in preventing monkeypox.
- Imvanex has been licensed in the united states to prevent monkeypox and smallpox.
- Acam2000,which contains a live vaccinia virus, is licensed for immunization in people who are at least 18 years old and at high risk for smallpox infection. It can be used in people exposed to monkeypox if used under an expanded access investigational protocol.

COLLEGE OF MEDICINE

Treatment

- Currently there is no proven, safe treatment for monkeypox virus infection.

also supportiv

- An antiviral agent known as TECOVIRIMAT [ST-246] developed for smallpox was licensed by the European medical association for monkeypox in 2022.
- Data is not available on the effectiveness of cidofovir in treating human cases of monkeypox . However, both have proven activity against poxviruses in vitro and animal studies.
- For purposes of controlling a monkeypox outbreak, smallpox vaccine, antivirals and vaccinia immune globulin can be used.

Recommended public health action

Health care facilities to keep heightened suspicion in people who:

* Present with otherwise unexplained rash.
* Who have travelled in the last 21 days to a country that has recently confirmed or suspected cases of monkeypox.
* Report contact with a person or people with confirmed or suspected monkeypox.

COLLEGE OF MEDICINE

REFERENCES

1. Interim advisory for IDSP in view of monkeypox cases reported from few countries.
2. https://www.who.int/news-room/fact-sheets/detail/monkeypox.
3. www.cdc.gov/poxvirus/monkeypox.

Chicken pox

also known as varicella, is a highly contagious infectious disease caused by the varicella-zoster virus (VZV).

It primarily affects children but can occur at any age. The disease is characterized by a vesicular rash, fever, and systemic symptoms, progressing through stages of macules, papules, vesicles, and crusts.

Chickenpox is usually self-limiting but can lead to severe complications.

Global and Regional Prevalence

Chickenpox is endemic worldwide.

Most cases occur in temperate regions during late winter and spring.

Developed Countries:

Decreased incidence due to widespread vaccination programs (e.g., the USA, Europe).

Outbreaks still occur in unvaccinated groups.

Developing Countries:

Higher incidence due to limited access to vaccines.

Often underreported or misdiagnosed in rural areas..

Mode of Transmission

impo

Respiratory Droplets:

Droplet : short distance

Airborne : carried by air for long time and place

* Spread through coughing, sneezing, or talking.

* Direct Contact: Contact with fluid from vesicular lesions.

* Fomites: Indirect spread via contaminated surfaces (less common).

Infectious Period:

1–2 days before the rash appears until all lesions have crusted over.

Environmental Factors:

Highly contagious in crowded or poorly ventilated settings.

Clinical Features of Chickenpox

1. Incubation Period

The incubation period for chickenpox is typically 10–21 days, with an average of 14–16 days after exposure to the varicella-zoster virus.

During this period, the virus replicates in the upper respiratory tract and lymph nodes before viremia leads to the development of symptom

Signs and Symptoms

Prodromal Symptoms (1–2 days before rash)

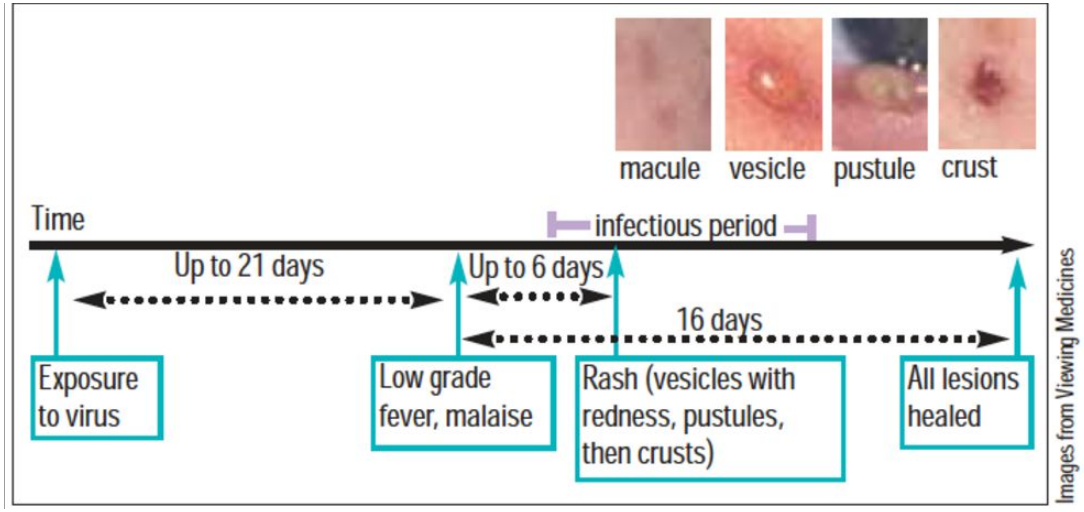
Fever (low-grade to moderate).

Fatigue and malaise, loss of appetite, Headache.

These symptoms are more pronounced in adults than in children.

Characteristic Rash:

Begins 1–2 days after the prodromal phase.



Key Differences between chicken pox and shingle

Primary vs. Reactivation:

Chickenpox results from the first encounter with the virus, while shingles is a reactivation of dormant VZV in nerve ganglia.

Rash: Chickenpox has a generalized rash, whereas shingles is localized to a single dermatome.

Contagiousness: Chickenpox is highly contagious; shingles is less so but can cause chickenpox in susceptible individuals.

Associated Pain: Pain is mild in chickenpox and much more severe in shingles, sometimes persisting as postherpetic neuralgia.

Laboratory Diagnosis

usuall clinical diagnose

Laboratory tests are used in atypical cases, severe disease,

or in immunocompromised patients where the clinical picture may be unclear.

Direct Tests:

Polymerase Chain Reaction (PCR),highly sensitive and specific.

Detects varicella-zoster virus DNA from vesicular fluid, blood, or swabs.

Direct Fluorescent Antibody (DFA) Test, rapidly identifies VZV in vesicle samples(less sensitive than PCR)

Prevention of Varicella Zoster (Chickenpox)

1. **Vaccination:** is the most effective method to prevent varicella-zoster infection.

Varicella Vaccine:

Type: **Live attenuated vaccine.**

Children: First dose at 12–15 months, Second dose at 4–6 years.

Adolescents and Adults (unvaccinated or non-immune):Two doses at least 4–8 weeks apart.

Efficacy: Provides immunity to >90% of recipients after two doses.

Herpes Zoster (Shingles) Vaccine:

Recommended for individuals >50 years to prevent reactivation of latent VZV (shingles).

COLLEGE OF MEDICINE

2. Isolation and Quarantine

→ have confirmed case at hospital

→ suspection at home

2. Isolation and Quarantine

Infected Individuals:

Isolate until all lesions have crusted (usually 7–10 days after rash onset).

Exposed Individuals (Non-immune):

Avoid contact with susceptible populations (pregnant women, neonates, immunocompromised individuals).

COLLEGE OF MEDICINE

3. Post-Exposure Prophylaxis

Varicella Vaccine:

Administered within 72 hours of exposure to prevent or reduce disease severity.

Varicella-Zoster Immune Globulin (VZIG):

usually contra-indication for live attenuated especially in 3rd trimester because of risk of congenital anomaly

For high-risk individuals who cannot receive the vaccine, such as, pregnant women, Immunocompromised individuals and Newborns exposed to maternal varicella.

COLLEGE OF MEDICINE

Clinical Implications

* If given within 96 hours:

VZIG is most likely to prevent or significantly reduce the severity of chickenpox.

* If given after 96 hours but before 10 days: It may still have some protective effect, especially in high-risk individuals, but the benefit diminishes.

* After 10 days: VZIG is unlikely to alter the course of the infection significantly, and close monitoring or antiviral treatment (e.g., acyclovir) may be required instead.

COLLEGE OF MEDICINE

imp

4. Hygiene and Environmental Measures

Hand hygiene:

Regular washing to reduce transmission.

for you and patient

Respiratory precautions: Use masks and maintain adequate ventilation in health care or crowded settings.

5. Public Health Strategies Promote universal childhood vaccination.

Conduct awareness campaigns about vaccine importance and disease severity.

Implement outbreak surveillance and control measures in schools and communities.

COLLEGE OF MEDICINE

DIFFERENCE BETWEEN THE "POXES"

	Monkeypox	**Smallpox**	**Chickenpox**
Species	Monkeypox Virus	Variola Virus	Varicella-zoster Virus
Symptoms	• Fever • Rash that starts from face to all parts of the body • Pus-filled boils which eventually forms scabs and fall off • Swollen lymph nodes	• Fever • Small red spots on tongue and mouth • Rash that spreads from face to other parts of the body • Pus-filled boils which eventually forms scabs and fall off	• Fever, Fatigues, Headaches • Rashes that forms itchy blisters • Frequently appearing on chest, back and face before spreading over the body • Blisters that are absent on palm and sole
Incubation	5-21 days	7-19 days	10-21 days
Mortality	1~10% depending on strain	Up to 30% depending on Type	Rare

COLLEGE OF MEDICINE

References

1. Centers for Disease Control and Prevention (CDC). Chickenpox (Varicella) Vaccination. Available at: https://www.cdc.gov/vaccines/vpd/varicella/index.html. Published November 22, 2016. Accessed December 10, 2024.
2. American Academy of Pediatrics (AAP). Postexposure Management and Vaccination for Varicella-Zoster Virus. In: Kimberlin DW, Barnett ED, Lynfield R, Sawyer MH, eds. Red Book: 2021–2024 Report of the Committee on Infectious Diseases. Itasca, IL: American Academy of Pediatrics; 2021: 831–842.
3. Dooling K, Marin M, Gershon AA. Clinical Manifestations of Varicella: Disease Is Largely Forgotten, but It's Not Gone. Journal of Infectious Diseases. 2022;226(Suppl 4):S380–S384. DOI:10.1093/infdis/jiac201.